Queen Mary
University of London

EBU5376
Digital Signal Processing

Continuous-Time Transformation

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Outline

- Time Domain vs. Frequency Domain
- Continuous-Time Fourier Series (CTFS)
- Continuous-Time Fourier Transform (CTFT)
- Laplace Transform

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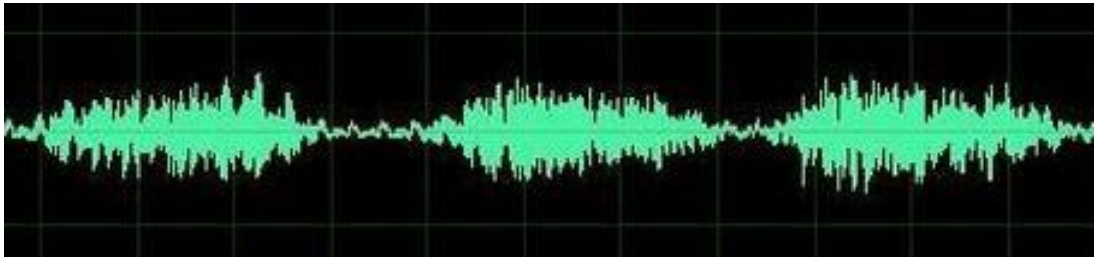
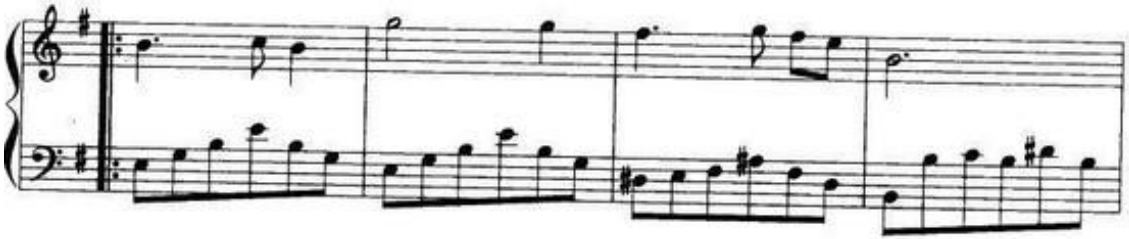
Time Domain vs. Frequency Domain

- Time-domain Representation:



- Frequency-time Representation:



Time Domain vs. Frequency Domain

- In the context of music, each note corresponds to a specific frequency
- Mathematically, they can be represented as pure sinusoids
- For instance, the frequency of note G4 is 392Hz, and is expressed as

$$x(t) = A\sin(392 \cdot 2\pi t)$$

- Music chords can be represented as a summation of pure sinusoids of different frequencies.

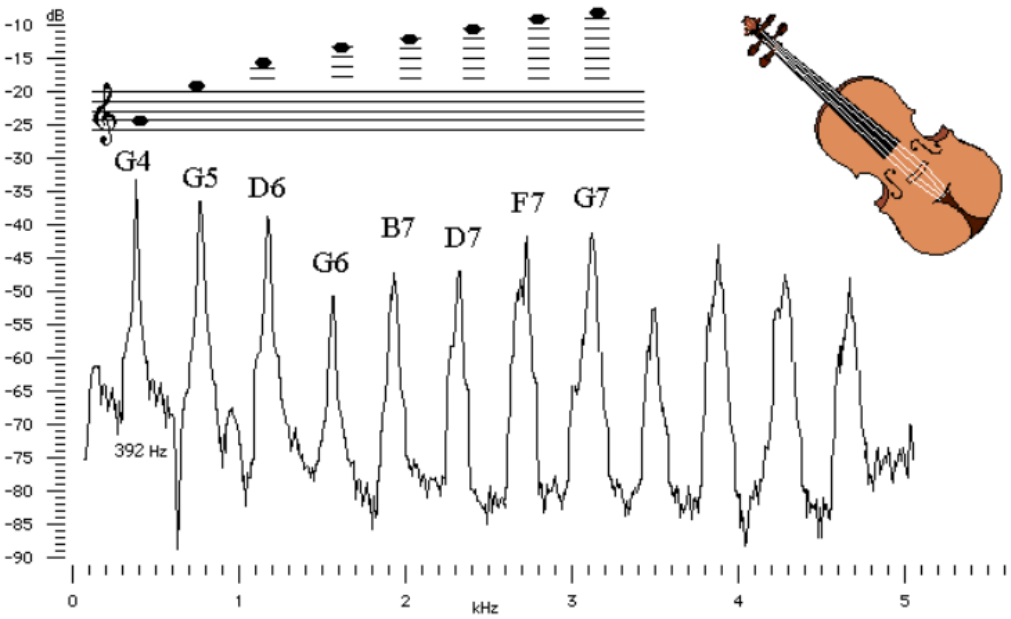
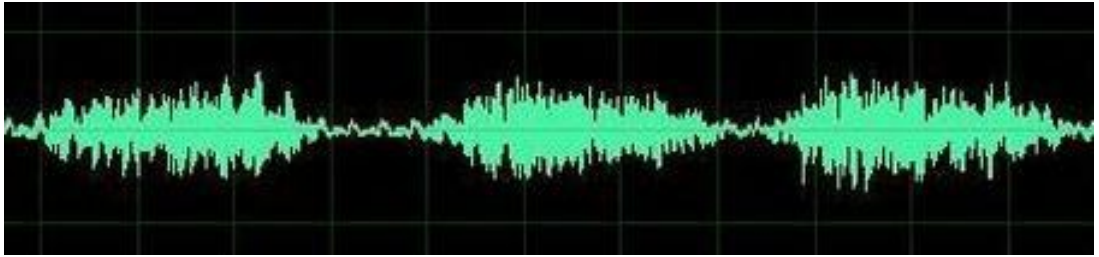


Fig: Frequencies of different violin notes.

Time Domain vs. Frequency Domain

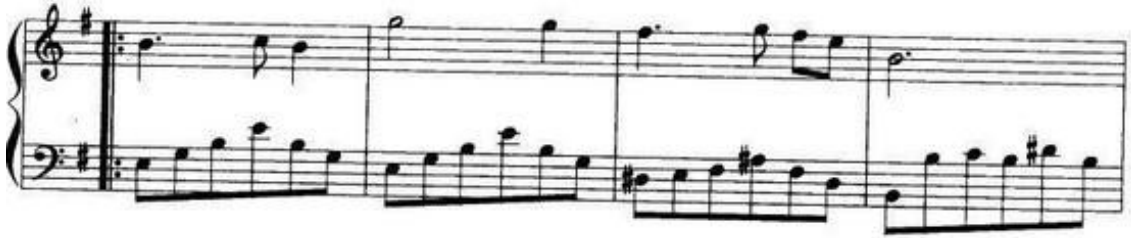
- Time-domain Representation:



- What information can you extract from the time-domain representation?
  - Amplitude (quiet or loud)
  - Signal duration
  - Number of peaks
  - …

Time Domain vs. Frequency Domain

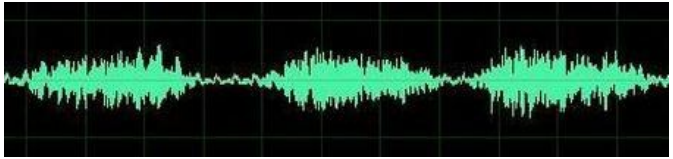
- Frequency-time Representation:

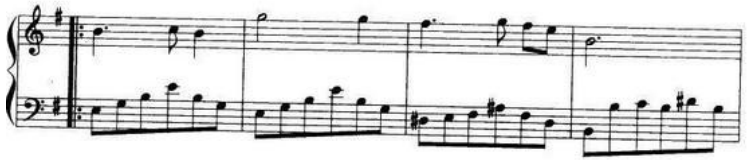


- What information can you extract from the music sheet?
  - Individual music notes
  - Spread of the notes
  - …
- More generally, in terms of DSP, these are the frequency-domain representations:
  - ✓ Frequency components
  - ✓ Frequency distributions

Time Domain vs. Frequency Domain

- Real-world signals (e.g., music pieces) can be equivalently presented in two fundamental domains: time and frequency domains.
- Each domain provides distinct insights into the signal, enabling diverse options for signal processing
- However, real-world signals often come in their time-domain representations.
- Obtaining accurate and informative frequency representations is a fundamental pursuit in DSP





Outline

- Time Domain vs. Frequency Domain
- Continuous-Time Fourier Series (CTFS)
- Continuous-Time Fourier Transform (CTFT)
- Laplace Transform

**For simplicity, the term 'Fourier Series' refers to the continuous-time Fourier series in this lecture.*

Continuous-Time Fourier Series (CTFS)

- Definition: The Fourier Series is an expansion of a periodic continuous-time function $x(t)$ in terms of an infinite weighted summation of sines and cosines, expressed as

$$x(t) = a_0 + \sum_{n=1}^{\infty} a_n \cos \omega n t + \sum_{n=1}^{\infty} b_n \sin \omega n t$$

  - T is the period of $x(t)$
  - $\omega = \frac{2\pi}{T}$ is the fundamental angular frequency of $x(t)$
  - $a_0, a_1, a_2, \ldots$ and $b_1, b_2, b_3, \ldots$ are the Fourier coefficients of $x(t)$
    - Why b_0 is not included?

- A **periodic signal** repeats every T seconds. To maintain this periodicity, only sine and cosine waves that also repeat every T or a fraction of T (like $T/2$, $T/3$) can combine to form the signal.

Continuous-Time Fourier Series (CTFS)

$$x(t) = a_0 + \sum_{n=1}^{\infty} a_n \cos \omega n t + \sum_{n=1}^{\infty} b_n \sin \omega n t$$

- The fundamental angular frequency $\omega = \frac{2\pi}{T}$ (rad/s) represents one full cycle of the signal within the period T (seconds).
- It is the smallest angular frequency at which the periodic signal repeats itself. Its multiples $(2\omega, 3\omega, \ldots)$ correspond to harmonics with higher frequencies that complete 2, 3, or more complete cycles within T.
- Signals that are **aperiodic** do not repeat over time. While Fourier Series can approximate their behavior within a specific interval, they cannot provide a precise representation for the entire signal.

Continuous-Time Fourier Series (CTFS)

$$\boxed{x(t) = a_0 + \underbrace{\sum_{n=1}^{\infty} a_n \cos \omega n t}_{\text{Even part of } x(t)} + \underbrace{\sum_{n=1}^{\infty} b_n \sin \omega n t}_{\text{Odd part of } x(t)}}$$

- $a_0 = \frac{1}{T}\int_0^T x(t)dt$ is the <u>mean value</u>, also known as the zero frequency term, or the direct current (DC) term.
- $a_n = \frac{2}{T}\int_0^T x(t)\cos \omega n t \, dt$ are the <u>even coefficients</u>
- $b_n = \frac{2}{T}\int_0^T x(t)\sin \omega n t \, dt$ are the <u>odd coefficients</u>

Continuous-Time Fourier Series (CTFS)

$$\boxed{x(t) = a_0 + \underbrace{\sum_{n=1}^{\infty} a_n \cos \omega n t}_{\text{Even part of } x(t)} + \underbrace{\sum_{n=1}^{\infty} b_n \sin \omega n t}_{\text{Odd part of } x(t)}}$$

- Interpretation of a_n and b_n
  - Coefficients a_n and b_n represent the weight of frequency ωn in the signal
  - If $a_n = b_n = 0$, it means the frequency ωn is absent in this signal
- Fourier Series of even and odd signals
  - If $x(t)$ is an even signal, then $b_n = 0, \forall n$
  - If $x(t)$ is an odd signal, then $a_n = 0, \forall n$ (including a_0)

Frequency Spectrum

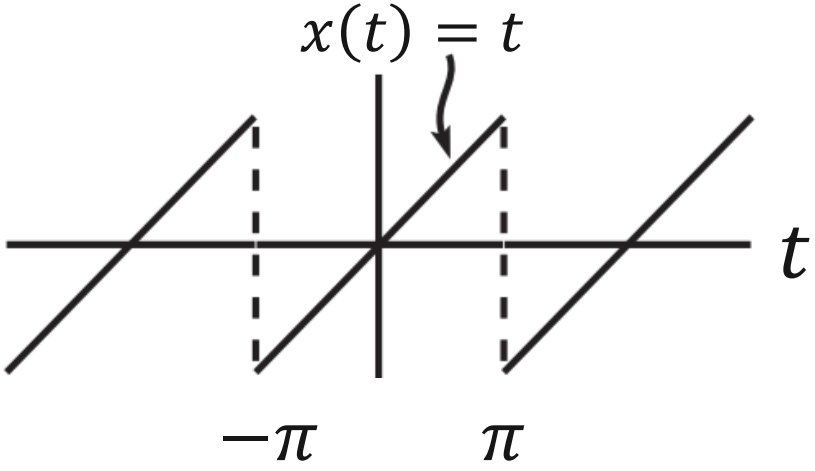
<u>What is a Frequency Spectrum?</u>

- The frequency spectrum represents a signal in terms of its component frequencies (horizontal axis) and their corresponding amplitudes (vertical axis).
- It describes how the signal's energy is distributed across different frequencies.
- Depending on the characteristics of the signal and the choice of transform method, the frequency spectrum can be continuous/discrete, periodic/aperiodic.

If a signal is represented using continuous-time Fourier series, what would the frequency spectrum look like?

Fourier Series – Example

- Derive the Fourier Series expansion of the Sawtooth function $x(t) = t$ defined over $(-\pi, \pi)$, shown below. Then, plot its frequency spectrum based on the Fourier coefficients.



Fourier Series of Sawtooth Function

The Fourier Series expansion of the Sawtooth function is

$$x(t) = \sum_{n=1}^{\infty} \frac{2}{n}(-1)^{n+1} \sin nt = 2\sin(t) - \sin 2t + \frac{2}{3}\sin(3t) - \frac{1}{2}\sin(4t) + \cdots$$

We see that

1. All harmonics (integer multiples of the fundamental frequency $\omega = 1$ rad/s) are present.
2. The amplitude of the n-th harmonic is proportional to $\frac{1}{n}$. Thus the amplitudes decay *linearly* by a factor of $\frac{1}{n}$.
3. There is an alternating sign, $(-1)^{n+1}$, which translates to a phase shift of 0 or π from term to term.

Graphically, if you plotted the **frequency spectrum**, it would be a set of **discrete lines** at frequencies $n\omega$ (with $n = 1,2, \ldots, \omega = 1$) whose heights are proportional to $\frac{1}{n}$.

Fourier Series of Sawtooth Function

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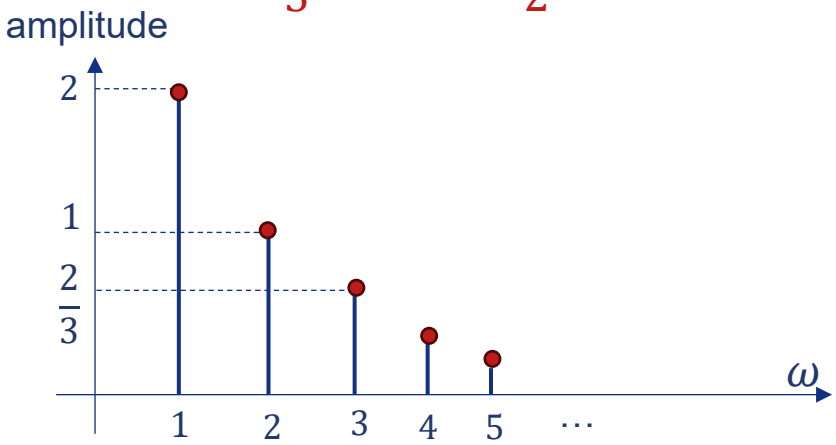
- The amplitude at frequency 1 rad/s is 2
- The amplitude at frequency 2 rad/s is 1
- The amplitude at frequency 3 rad/s is $\frac{2}{3}$
- The amplitude at frequency n rad/s is $\frac{2}{n}$
- …



- There are infinitely many non-zero amplitudes, corresponding to all integer multiples of the fundamental frequency
- What about the negative frequencies?

Fourier Series – Exponential Form

- The trigonometric Fourier series only describes positive frequencies, but negative frequencies emerge naturally when reformulated into the complex exponential form, given by

$$x(t) = \sum_{n=-\infty}^{\infty} C_n e^{j\omega n t}, \qquad C_n = \frac{1}{T}\int_0^T x(t) e^{-j\omega n t} dt$$

  - C_n: The complex Fourier coefficient for the n-th harmonic.
  - T: The period of the signal $x(t)$
  - $\omega = \frac{2\pi}{T}$: The fundamental angular frequency
  - n: An integer representing the harmonic index (can be positive, negative, or zero).
  - Full derivations from the trigonometric form to the exponential form is in Appendix B

Exponential vs. Trigonometric Fourier Series

- The **Exponential Fourier Series** represents a signal using complex exponentials, with coefficients that are generally complex-valued.
- When the input signal is real-valued, the coefficients exhibit conjugate symmetry ($c_{-n} = c_n^*$), ensuring the imaginary parts cancel out in the time-domain reconstruction. Hence, the frequency spectrum of real-valued signals is symmetric about the vertical axis.
- The **Trigonometric Fourier Series** uses real-valued coefficients a_n (cosine terms) and b_n (sine terms).
- If the input signal is real-valued, these coefficients are always real.
- For complex-valued input signals, the coefficients a_n and b_n can also become complex, though in such cases, the exponential Fourier series is typically preferred because it naturally accommodates complex signals.

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**For simplicity, the term 'Fourier Transform' refers to the continuous-time Fourier transform in this lecture.*

Fourier Transform

- Fourier Series can fully represent periodic signals. However, periodic signals are non-informational.
- Aperiodic signals (signals containing information) cannot be represented using the Fourier Series, the **Fourier Transform (FT)** is required.
- From FS to FT: An aperiodic signal can be thought of a periodic signal with an **infinitely long period $\boldsymbol{T}$**. As $T \to \infty$, the discrete frequency spectrum of the FS becomes continuous, leading to the development of the Fourier Transform.
- The FT provides a unified framework to analyze the frequency spectrum of both periodic and aperiodic signals.

Fourier Series vs. Fourier Transform

- Using **Fourier Series**, the signal is represented as

$$x(t) = \sum_{n=-\infty}^{\infty} C_n e^{j\omega n t}$$

- Using **Fourier Transform**, the signal is represented as

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega$$

  - $X(\omega)$ is called the Fourier Transform of $x(t)$, calculated as

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

- We denote the Fourier transform pair of a signal $x(t)$ as

$$x(t) \overset{\mathcal{F}}{\leftrightarrow} X(\omega), \qquad where\ \mathcal{F}\big(x(t)\big) = X(\omega), \qquad x(t) = \mathcal{F}^{-1}(X(\omega))$$

Fourier Transform

- The Fourier Transform $X(\omega)$ is a complex-valued function.

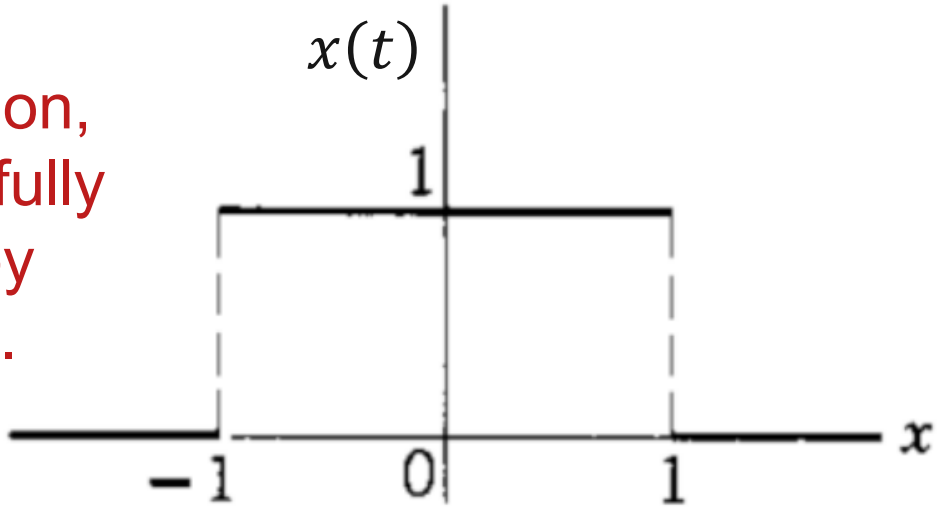
$$X(\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}dt = \int_{-\infty}^{\infty} x(t)\,(\cos\omega t - j\sin\omega t)\,dt$$

  - The real part $Re\big(X(\omega)\big)$ represents the contribution of the **cosine waveforms** at frequency ω
  - The imaginary part $Im\big(X(\omega)\big)$ represents the contribution of the **sine waveforms** at frequency ω
  - The magnitude $|X(\omega)|$ indicates the overall amplitude of the frequency component at ω
  - The phase $\angle X(\omega) = \tan^{-1}\left(\frac{Im\big(X(\omega)\big)}{Re\big(X(\omega)\big)}\right)$ describes the relative shift of the frequency component

Fourier Transform Example

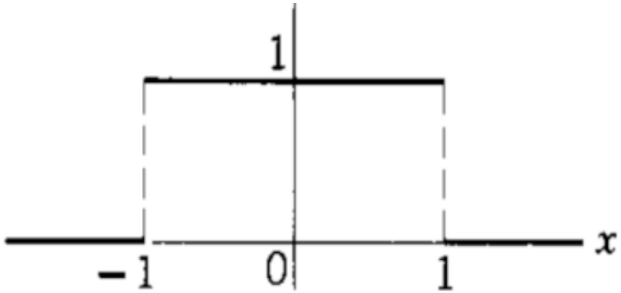
➢ <u>Example</u>: Find the Fourier transform of $x(t) = \begin{cases} 1, & -1 < t < 1 \\ 0, & |t| > 1 \end{cases}$

This is not a periodic function, so cannot be fully represented by Fourier series.



Fourier Transform Example

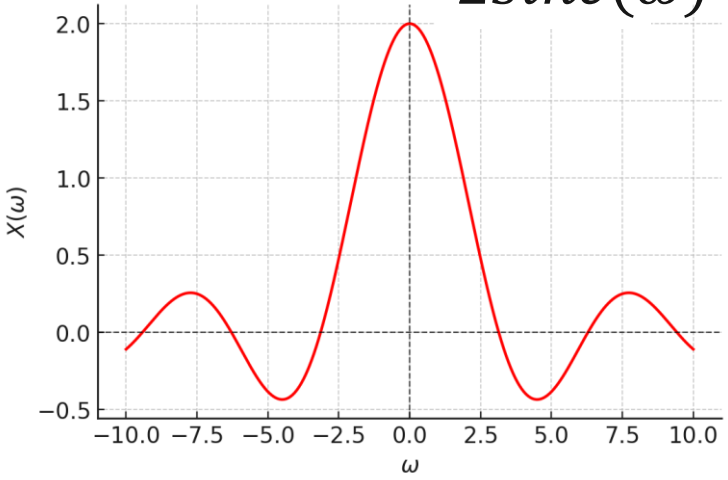
$x(t)$



Time Domain

Fourier Transform

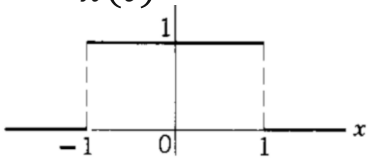
$2sinc(\omega)$



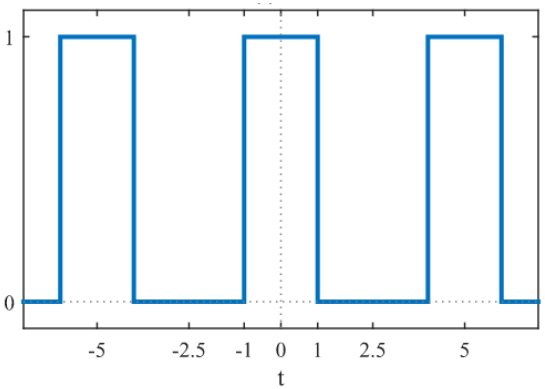
Frequency Domain

Fourier Transform of Periodic Signals

$x(t)$

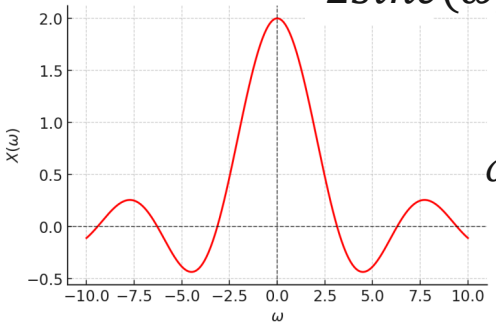


periodize



Fourier Transform

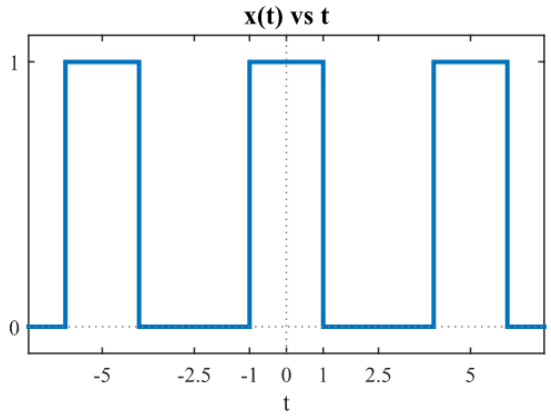
$2sinc(\omega)$



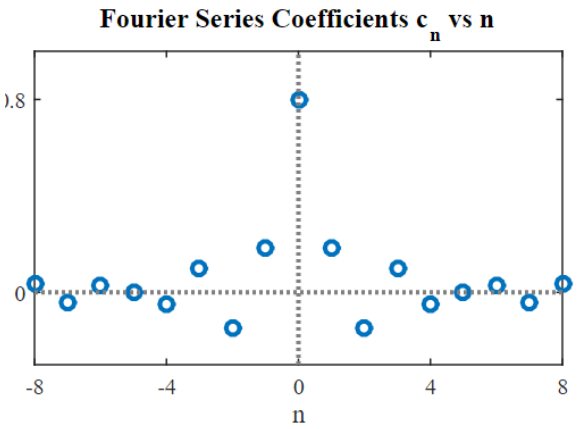
ω

Fourier Transform

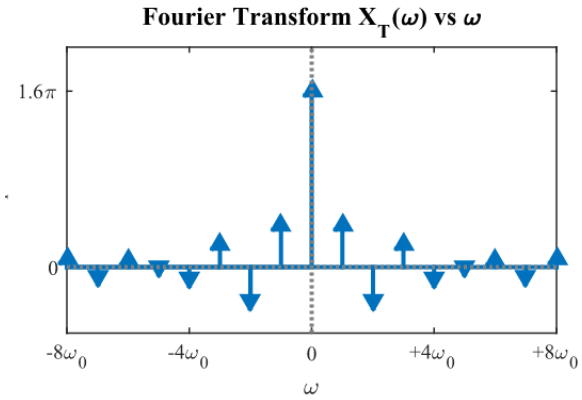
Fourier Transform of Periodic Signals



$x(t)$



Fourier Series



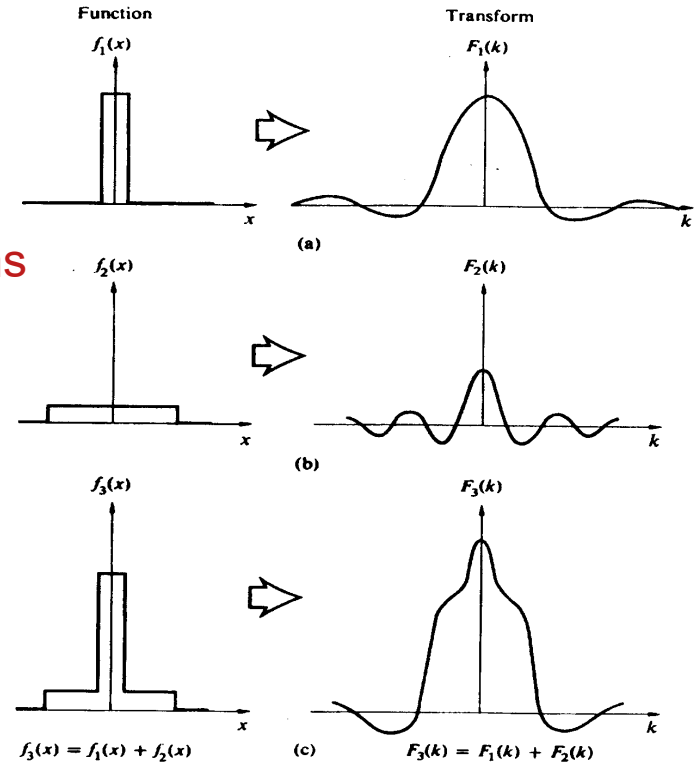
Fourier Transform

- The FT of a periodic signal consists of impulses at discrete harmonic frequencies $n\omega$, with magnitudes equal to the FS coefficients.

Properties of Fourier Transform

- The Fourier Transform exhibits many useful properties.
- For example:
  1. **Linearity**
  2. Time-shift
  3. Frequency-shift
  4. Time scaling
  5. Time differentiation
  6. Time integration
  7. Convolution
- Formulations and proofs of these properties can be found in Appendix C

The Fourier Transform of a sum of two functions is the sum of the Fourier Transforms of each function



Properties of Fourier Transform

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  4. **Time scaling**
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  7. Convolution
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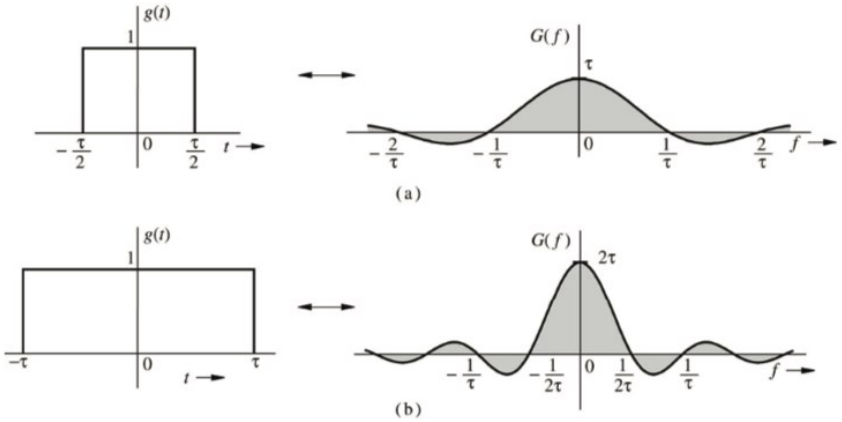
Reducing the time-domain duration of a waveform causes its spectrum to spread out in the frequency domain, dictating a larger system bandwidth.



Properties of Fourier Transform

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- For example:
  1. Linearity
  2. Time-shift
  3. Frequency-shift
  4. Time scaling
  5. Time differentiation
  6. Time integration
  7. **Convolution**
- Formulations and proofs of these properties can be found in Appendix C

Let $x_1(t)$ and $x_2(t)$ be functions with Fourier Transforms $X_1(\omega)$ and $X_2(\omega)$

The convolution theorem states that:

$$\mathcal{F}\{x_1(t) * x_2(t)\} = X_1(\omega) \cdot X_2(\omega)$$

and

$$\mathcal{F}^{-1}\{X_1(\omega) * X_2(\omega)\} = x_1(t) \cdot x_2(t)$$

In other words, convolution in the time domain corresponds to multiplication in the frequency domain, and vice versa.

Fourier Transform of Periodic Signals

- The periodization operation is essentially a <u>convolution</u> of the signal $x(t)$ with a train of Dirac delta functions:

$$x_{periodic}(t) = x(t) * \sum_{n=-\infty}^{\infty} \delta(t - nT_0) = \sum_{n=-\infty}^{\infty} x(t - nT_0)$$

  - This shifts $x(t)$ by multiples of T_0 (both forward and backward in time), and sums all the shifted versions, resulting in a periodic signal with period T_0

- According to the convolution theorem, the frequency spectrum of $x_{periodic}(t)$ can be derived as

$$\mathcal{F}\left\{x(t) * \sum_{n=-\infty}^{\infty} \delta(t - nT_0)\right\} = \mathcal{F}\{x(t)\} \cdot \mathcal{F}\left\{\sum_{n=-\infty}^{\infty} \delta(t - nT_0)\right\} = X(\omega) \cdot \frac{2\pi}{T_0} \sum_{k=-\infty}^{\infty} \delta(\omega - k\omega_0)$$

The derivations of the Fourier transform for the impulse train can be found in Appendix D

Fourier Transform of Periodic Signals

- By periodizing the signal with a period T_0, the spectrum becomes a series of impulses spaced by T_0
- This corresponds to sampling the spectrum at intervals determined by the repetition period T_0
- Periodization in Time ↔ Sampling in Frequency

$$x(t) * \sum_{n=-\infty}^{\infty} \delta(t - nT_0) \leftrightarrow \frac{2\pi}{T_0} \sum_{k=-\infty}^{\infty} X(\omega)\delta(\omega - k\omega_0)$$

where $\omega_0 = \frac{2\pi}{T_0}$ is the fundamental angular frequency

Fourier Transform of Periodic Signals

- As a result of sampling, the frequency spectrum now consists of impulses at $\omega = k\omega_0$, with amplitudes of $\frac{2\pi}{T_0}X(k\omega_0)$, where k is any integer
- The frequency spectrum of $x_{periodic}(t)$ should be consistent in both Fourier Series and Fourie transform representations.
- Let C_k denote the Fourier Series coefficients (exponential formula), hence

$$C_k = \frac{2\pi}{T_0}X(k\omega_0)$$

describes how the Fourier series coefficients relate to the sampled frequency spectrum

Application in 4G

- The Fourier Transform is crucial for modern communication systems, enabling Orthogonal Frequency Division Multiplexing (OFDM), the backbone of 4G communications
- OFDM exploits the fact that different sine waves (at different frequencies) are **orthogonal**, meaning they don't interfere if spaced correctly.
- Then, data is split into multiple sub-streams, each sent on a distinct frequency (**subcarrier**).
- The transmitter uses an Inverse Fourier Transform to combine subcarriers into one time-domain signal, and the receiver uses a Fourier Transform to recover each sub-stream without interference
- Because subcarriers are orthogonal, they can overlap in frequency yet still be separated, maximizing bandwidth usage.

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Laplace Transform

- Laplace transform is a continuous-time transformation method for solving linear differential equations and studying the behavior of analog systems.
- It involves transforming a time-domain signal into the complex frequency domain. The equation of Laplace transform is given by

$$X(s) = \mathcal{L}\big(x(t)\big) = \int_0^{\infty} x(t)e^{-st}\,dt$$

  - $s = \sigma + j\omega$ is a complex variable defined on the s-plane
  - The s-plane is a rectangular coordinate system with ω expressing the distance along the real (horizontal) axis, and σ the distance along the imaginary (vertical) axis.

Laplace Transform

unit impulse function

- <u>Example 1</u>: The Laplace transform of signal $\delta(t) = \begin{cases} 0, & t \neq 0 \\ \infty, & t = 0 \end{cases}$ is given by

unit step function

- <u>Example 2</u>: The Laplace transform of signal $u(t) = \begin{cases} 0, & t < 0 \\ 1, & t \geq 0 \end{cases}$ is given by

Relating Laplace Transform and CTFT

- There is a close relationship between Laplace Transform (LT) and Continuous-Time Fourier Transform (CTFT)

Laplace Transform: $$X(s) = \int_0^{\infty} x(t)e^{-(\sigma+j\omega)t}\,dt$$

CTFT: $$X(\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}dt$$

- CTFT can be viewed as a special case of LT, when $\sigma = 0$ and $t \geq 0$
- However, these two transforms are <u>distinct mathematical tools</u> with different purposes and mathematical foundations
  - CTFT is used to analyze the frequency components of signals
  - LT is used to analyze linear differential equations and linear systems

Summary – Continuous-Time Transformation

- Frequency spectrum provides a clear and precise visualization of the frequency components in the signal. Fourier transform is the main tool for obtaining the frequency spectrum.
- Depending on the nature of the signal, Fourier transform may take different forms:
  1. *Continuous-Time Fourier Series (CTFS)*
     - Can only represent periodic continuous-time signals
     - Produce a discrete frequency spectrum
  2. *Continuous-Time Fourier Transform (CTFT)*
     - Can represent both periodic and aperiodic continuous-time signals
     - Produce a continuous frequency spectrum
     - If applied to periodic signal, the spectrum will consist of evenly spaced impulses

Summary – Continuous-Time Transformation

- Periodization in Time ↔ Sampling in Frequency
- Laplace transform is also a continuous-time transformation method, but it is designed for solving linear differential equations and studying the behavior of analog systems.
  - CTFT and Laplace transform share a similar formula.
  - More particularly, CTFT can be viewed as a special case of Laplace transform

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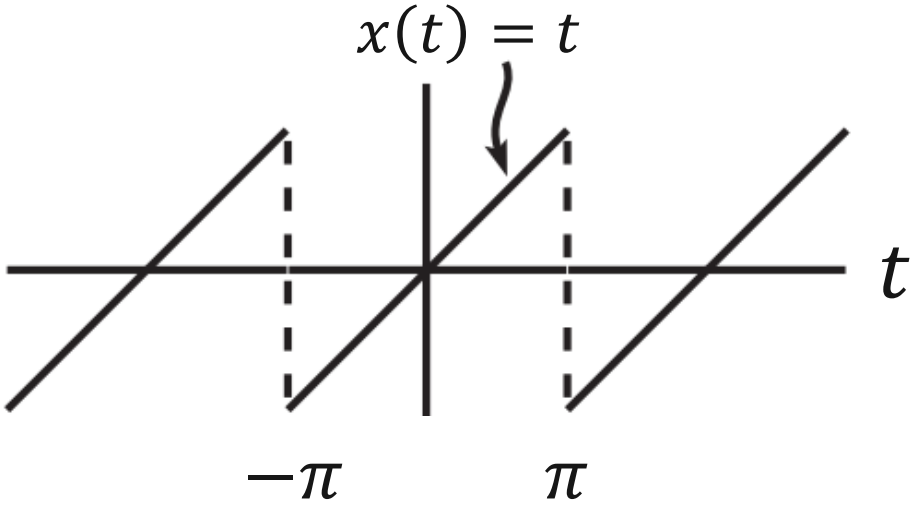
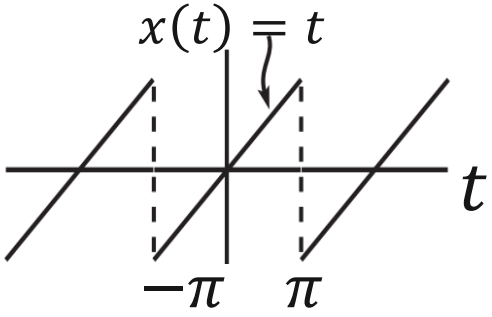
Appendix A - Fourier Series of Sawtooth

- Example: Compute the Fourier Series for the Sawtooth function $x(t) = t$ defined over $(-\pi, \pi)$, shown below



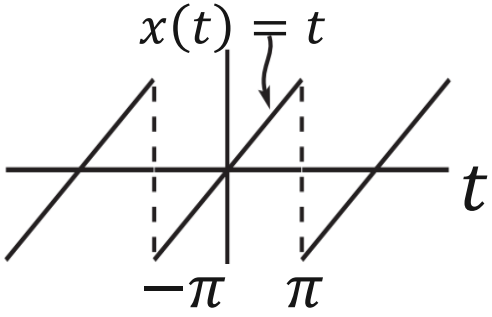
Appendix A - Fourier Series of Sawtooth



- Solution:
  - This waveform has expression $x(t) = t$, for $-\frac{\pi}{2} < t < \frac{\pi}{2}$, with a period of $T = 2\pi$
  - The fundamental angular frequency of $x(t)$ is $\omega = \frac{2\pi}{T} = 1$
  - We notice that this waveform is odd, hence, $a_n = 0, \forall n$, and we only need to calculate b_n
  - Thus, the Fourier series expansion of the signal is

$$x(t) = \sum_{n=1}^{\infty} b_n \sin nt$$

Appendix A - Fourier Series of Sawtooth



- Solution:
  - Since $T = 2\pi, \omega = 1,$ the odd coefficients b_n can be computed by

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} t \sin nt \, dt$$

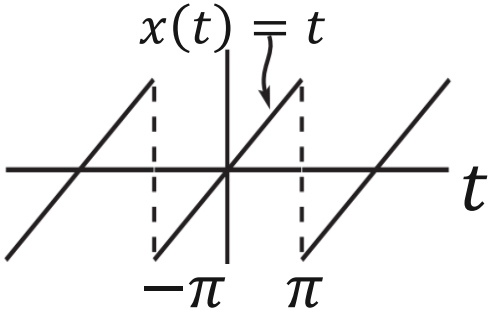
Using integration by parts, we let $u = t, dv = \sin nt \, dt$

Then, $du = dt, v = \int \quad \sin nt \, dt = -\frac{\cos nt}{n}$.

- So the integral becomes:

$$b_n = \frac{1}{\pi}\left[\left(-\frac{t\cos nt}{n}\right)\Big|_{t=-\pi}^{\pi} - \int_{-\pi}^{\pi}\left(-\frac{\cos nt}{n}\right)dt\right]$$

Appendix A - Fourier Series of Sawtooth



- Solution:
  - So the integral becomes:

$$b_n = \frac{1}{\pi}\left[\left(-\frac{t\cos nt}{n}\right)\Big|_{t=-\pi}^{\pi} - \int_{-\pi}^{\pi}\left(-\frac{\cos nt}{n}\right)dt\right]$$

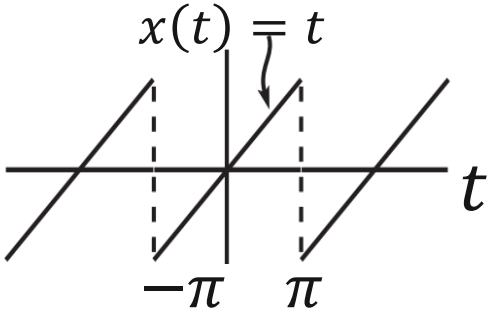
$$= \frac{1}{n\pi}\left[-2\pi\cos n\pi + \left(\frac{\sin nt}{n}\right)\Big|_{t=-\pi}^{\pi}\right]$$

$$= \frac{1}{n\pi}\left[-2\pi\cos n\pi + \frac{1}{n}(\sin n\pi - \sin(-n\pi))\right]$$

$$= \frac{1}{n\pi}\left[-2\pi\cos n\pi + \frac{2\sin n\pi}{n}\right]$$

$$= \frac{2}{n\pi}\left[\frac{\sin n\pi}{n} - \pi\cos n\pi\right]$$

Appendix A - Fourier Series of Sawtooth



- Solution:
  - From previous page, we have

$$b_n = \frac{2}{n\pi}\left[\frac{\sin n\pi}{n} - \pi\cos n\pi\right]$$

  - Since $\sin n\pi = 0$, $\cos n\pi = (-1)^n$ for all integer n
  - We have

$$b_n = \frac{2}{n\pi}(-\pi(-1)^n) = -\frac{2}{n}(-1)^n = \frac{2}{n}(-1)^{n+1}$$

  - Finally, the Fourier series of the sawtooth function is

$$x(t) = \sum_{n=1}^{\infty} \frac{2}{n}(-1)^{n+1}\sin nt$$

Appendix B – Exponential Fourier Series

Here, we show the derivation from Trigonometric Fourier Series (FS) to Exponential Fourier Series .

- The equation of the Trigonometric FS is

$$x(t) = a_0 + \sum_{n=1}^{\infty} a_n \cos \omega n t + \sum_{n=1}^{\infty} b_n \sin \omega n t$$

- We will use the following trigonometric properties:

$$\cos \omega t = \frac{e^{j\omega t} + e^{-j\omega t}}{2} \qquad \sin \omega t = \frac{e^{j\omega t} - e^{-j\omega t}}{j2}$$

Appendix B – Exponential Fourier Series

- By substituting the trig. equations into the Fourier series, we get

$$x(t) = a_0 + \sum_{n=1}^{\infty} a_n \frac{e^{j\omega nt} + e^{-j\omega nt}}{2} + \sum_{n=1}^{\infty} b_n \frac{e^{j\omega nt} - e^{-j\omega nt}}{2j}$$

- By collecting the coef. of the same terms, the equation becomes

$$x(t) = a_0 + \sum_{n=1}^{\infty} \left(\frac{a_n}{2} + \frac{b_n}{2j}\right) e^{j\omega nt} + \sum_{n=1}^{\infty} \left(\frac{a_n}{2} - \frac{b_n}{2j}\right) e^{-j\omega nt}$$

- So, we define a new set of coef. By

$$C_n = \frac{a_n}{2} - j\frac{b_n}{2}, n > 0; \qquad C_n = \frac{a_{-n}}{2} + j\frac{b_{-n}}{2}, n < 0; \qquad C_0 = a_0$$

Appendix B – Exponential Fourier Series

- Hence, the Fourier series becomes

$$x(t) = C_0 + \sum_{n=1}^{\infty} C_n e^{j\omega nt} + \sum_{n=1}^{\infty} C_{-n} e^{-j\omega nt}$$

$$= C_0 e^{j\omega 0t} + \sum_{n=1}^{\infty} C_n e^{j\omega nt} + \sum_{n=-\infty}^{-1} C_n e^{j\omega nt}$$

$$= \sum_{n=-\infty}^{\infty} C_n e^{j\omega nt}$$

This is the **exponential Fourier series**

Appendix B – Exponential Fourier Series

➢ Compute the exact form of C_n:

The formula of a_n and b_n are

$$a_n = \frac{2}{T}\int_0^T x(t)\cos\omega nt\,dt$$

$$b_n = \frac{2}{T}\int_0^T x(t)\sin\omega nt\,dt$$

Using the previous results

$$C_n = \frac{a_n}{2} - j\frac{b_n}{2}, n \geq 0; \qquad C_n = \frac{a_{-n}}{2} + j\frac{b_{-n}}{2}, n < 0$$

The formula of C_n is derived as

$$C_n = \frac{1}{T}\int_0^T x(t)\cos\omega nt\,dt - j\,\frac{1}{T}\int_0^T x(t)\sin\omega nt\,dt\,, \qquad n \geq 0$$

$$C_n = \frac{1}{T}\int_0^T x(t)\cos\omega nt\,dt + j\,\frac{1}{T}\int_0^T x(t)\sin\omega nt\,dt\,, \qquad n < 0$$

Appendix B – Exponential Fourier Series

➢ Compute the exact form of C_n:

The result of the previous page can be simplified to

$$C_n = \frac{1}{T}\int_0^T x(t)\,(\cos\omega nt - j\sin\omega nt)dt = \frac{1}{T}\int_0^T x(t)e^{-j\omega nt}dt\,, \qquad n \geq 0$$

$$C_n = \frac{1}{T}\int_0^T x(t)\,(\cos\omega nt + j\sin\omega nt)dt = \frac{1}{T}\int_0^T x(t)e^{j\omega nt}dt\,, \qquad n < 0$$

These two expressions can be combined into

$$C_n = \frac{1}{T}\int_0^T x(t)e^{-j\omega nt}dt$$

That is the exact formula for the Fourier coefficient C_n

Appendix C – Properties of Fourier Transform

➢ Linearity

❖ <u>Definition</u>: Suppose $x_1(t) \overset{\mathcal{F}}{\leftrightarrow} X_1(\omega)$ and $x_2(t) \overset{\mathcal{F}}{\leftrightarrow} X_2(\omega)$, then for any real or complex scalars a and b, we have

$$ax_1(t) + bx_2(t) \overset{\mathcal{F}}{\leftrightarrow} aX_1(\omega) + bX_2(\omega)$$

❖ <u>Proof</u>: $ax_1(t) + bx_2(t) \overset{\mathcal{F}}{\leftrightarrow} \int_{-\infty}^{\infty} x(t)e^{-j\omega t}dt = \int_{-\infty}^{\infty} \big(ax_1(t) + bx_2(t)\big)e^{-j\omega t}dt$

$$= a\int_{-\infty}^{\infty} x_1(t)e^{-j\omega t}dt + b\int_{-\infty}^{\infty} x_2(t)e^{-j\omega t}dt = aX_1(\omega) + bX_2(\omega)$$

Appendix C – Properties of Fourier Transform

➢ Time shift

❖<u>Definition</u>: Suppose $x(t) \overset{\mathcal{F}}{\leftrightarrow} X(\omega)$, then for any real number a, we have

$$x(t-a) \overset{\mathcal{F}}{\leftrightarrow} X(\omega)e^{-j\omega a}$$

❖<u>Proof</u>: $x(t-a) \overset{\mathcal{F}}{\leftrightarrow} \int_{-\infty}^{\infty} x(t-a)e^{-j\omega t}dt$

$$= \int_{-\infty}^{\infty} x(t')e^{-j\omega(t'+a)}dt'$$

$$= e^{-j\omega a} \int_{-\infty}^{\infty} x(t')e^{-j\omega t'}dt' = X(\omega)e^{-j\omega a}$$

Appendix C – Properties of Fourier Transform

➢ Time scaling

❖ <u>Definition</u>: Suppose $x(t) \overset{\mathcal{F}}{\leftrightarrow} X(\omega)$, then for any positive real number a, we have

$$x(at) \overset{\mathcal{F}}{\leftrightarrow} \frac{1}{a} X\left(\frac{\omega}{a}\right)$$

❖ <u>Proof</u>: $x(at) \overset{\mathcal{F}}{\leftrightarrow} \int_{-\infty}^{\infty} x(at) e^{-j\omega t} dt = \int_{-\infty}^{\infty} x(t') e^{-j\omega\left(\frac{t'}{a}\right)} d\left(\frac{t'}{a}\right)$

$$= \frac{1}{a} \int_{-\infty}^{\infty} x(t') e^{-j\frac{\omega}{a}t'} dt' = \frac{1}{a} X\left(\frac{\omega}{a}\right)$$

Appendix C – Properties of Fourier Transform

➢ Frequency shift

❖ <u>Definition</u>: Suppose $x(t) \overset{\mathcal{F}}{\leftrightarrow} X(\omega)$, then for some ω_0, we have

$$e^{j\omega_0 t}x(t) \overset{\mathcal{F}}{\leftrightarrow} X(\omega - \omega_0)$$

❖ <u>Proof</u>: $e^{j\omega_0 t}x(t) \overset{\mathcal{F}}{\leftrightarrow} \int_{-\infty}^{\infty} x(t)e^{-j\omega t}e^{j\omega_0 t}dt$

$$= \int_{-\infty}^{\infty} x(t)e^{-j(\omega-\omega_0)t}dt = X(\omega - \omega_0)$$

Appendix C – Properties of Fourier Transform

➢ Time differentiation

❖ <u>Definition</u>: Suppose $x(t) \overset{\mathcal{F}}{\leftrightarrow} X(\omega)$, then

$$\frac{dx(t)}{dt} \overset{\mathcal{F}}{\leftrightarrow} j\omega X(\omega)$$

❖ <u>Proof</u>: $\frac{dx(t)}{dt} = \frac{d}{dt}\left(\frac{1}{2\pi}\int_{-\infty}^{\infty} X(\omega)e^{j\omega t}d\omega\right) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\omega)\frac{d}{dt}(e^{j\omega t})d\omega$

$$= j\omega \times \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\omega)e^{j\omega t}d\omega = j\omega \times x(t) \overset{\mathcal{F}}{\leftrightarrow} j\omega X(\omega)$$

Appendix C – Properties of Fourier Transform

➢ Time integration

❖ <u>Definition</u>: Suppose $x(t) \overset{\mathcal{F}}{\leftrightarrow} X(\omega)$, then

$$\int_{-\infty}^{t} x(\lambda)d\lambda \overset{\mathcal{F}}{\leftrightarrow} \frac{1}{j\omega}X(\omega) + \pi X(0)\delta(\omega)$$

❖ If the signal has no direct current (DC) component, i.e., $X(0) = 0$, then

$$\int_{-\infty}^{t} x(\lambda)d\lambda \overset{\mathcal{F}}{\leftrightarrow} \frac{1}{j\omega}X(\omega)$$

❖ This means the integration of a time function has the effect of dividing its Fourier transform by the factor of $j\omega$.

Appendix C – Properties of Fourier Transform

➢ Convolution in time

❖ <u>Definition</u>: Suppose $x_1(t) \overset{\mathcal{F}}{\leftrightarrow} X_1(\omega)$ and $x_2(t) \overset{\mathcal{F}}{\leftrightarrow} X_2(\omega)$, then

$$x_1(t) * x_2(t) \overset{\mathcal{F}}{\leftrightarrow} X_1(\omega)X_2(\omega)$$

❖ Continuous-time convolution is defined as

$$x_1(t) * x_2(t) = \int_{-\infty}^{\infty} x_1(\lambda)x_2(t-\lambda)d\lambda$$

❖ This property means that convolution in the time domain corresponds to multiplication in the frequency domain.

Appendix C – Properties of Fourier Transform

➢ Convolution in time (proof)

$$x_1(t) * x_2(t) = \int_{\lambda=-\infty}^{\lambda=\infty} x_1(\lambda) x_2(t-\lambda) d\lambda$$

$$\overset{\mathcal{F}}{\leftrightarrow} \int_{t=-\infty}^{t=\infty} \int_{\lambda=-\infty}^{\lambda=\infty} x_1(\lambda) x_2(t-\lambda) e^{-j\omega t} d\lambda \, dt$$

← Perform Fourier transform

$$= \int_{\lambda=-\infty}^{\lambda=\infty} x_1(\lambda) \left(\int_{t=-\infty}^{t=\infty} x_2(t-\lambda) e^{-j\omega t} dt \right) d\lambda$$

← Rearranging terms

$$= \int_{\lambda=-\infty}^{\lambda=\infty} x_1(\lambda) \left(\int_{t=-\infty}^{t=\infty} x_2(u) e^{-j\omega(\lambda+u)} du \right) d\lambda$$

← Substitute $u = t - \lambda$

$$= \left(\int_{\lambda=-\infty}^{\lambda=\infty} x_1(\lambda) \, e^{-j\omega\lambda} d\lambda \right) \left(\int_{t=-\infty}^{t=\infty} x_2(u) e^{-j\omega u} du \right) = X_1(\omega) X_2(\omega)$$

Appendix D - Fourier Transform of Impulse Train

- Let $X(\omega)$ denotes the Fourier transform of $x(t)$. Using the convolution theorem, derive the Fourier transform of $x_{periodic}(t)$.

$$x_{periodic}(t) = x(t) * \sum_{n=-\infty}^{\infty} \delta(t - nT_0)$$

Convolution Theorem: Convolution in the time domain corresponds to multiplication in the frequency domain.

So $\mathcal{F}\{x_{periodic}(t)\} = \mathcal{F}\{x(t) * \sum_{n=-\infty}^{\infty} \delta(t - nT_0)\} = \mathcal{F}\{x(t)\} \cdot \mathcal{F}\{\sum_{n=-\infty}^{\infty} \delta(t - nT_0)\}$

$$\mathcal{F}\left\{\sum_{n=-\infty}^{\infty} \delta(t - nT_0)\right\} = \int_{-\infty}^{\infty} \sum_{n=-\infty}^{\infty} \delta(t - nT_0)\, e^{-j\omega t} dt = \sum_{n=-\infty}^{\infty} \int_{-\infty}^{\infty} \delta(t - nT_0) e^{-j\omega t} dt$$

$$= \sum_{n=-\infty}^{\infty} e^{-j\omega n T_0}$$

Since $\delta(t - nT_0)$ is nonzero only at $t = nT_0$, each integral collapses to the value of the exponential at $t = nT_0$:

Appendix D - Fourier Transform of Impulse Train

So far we obtained

$$\mathcal{F}\left\{\sum_{n=-\infty}^{\infty} \delta(t - nT_0)\right\} = \sum_{n=-\infty}^{\infty} e^{-j\omega n T_0}$$

The above spectrum is periodic in ω with period $\omega_0 = \frac{2\pi}{T_0}$

Also, a well-known (and very useful) fact is that this infinite sum can be represented by an infinite sum of Dirac delta functions in the ω domain. One way to see this is via the **Poisson summation formula**, which—roughly speaking—says that summing samples of a function in the time domain corresponds to summing impulses (a “Dirac comb”) in the frequency domain (and vice versa).

A commonly cited result is:

$$\sum_{n=-\infty}^{\infty} e^{-j\alpha n} = 2\pi \sum_{k=-\infty}^{\infty} \delta(\alpha - 2\pi k)$$

In our problem, $\alpha = \omega T_0$

Appendix D - Fourier Transform of Impulse Train

Then,

$$\sum_{n=-\infty}^{\infty} e^{-j\omega n T_0} = 2\pi \sum_{k=-\infty}^{\infty} \delta(\omega T_0 - 2\pi k)$$

We can rewrite the delta argument $\omega T_0 - 2\pi k = 0$ as $\omega - k\left(\frac{2\pi}{T_0}\right) = 0$. Note that factoring out T_0 from inside the delta then introduces a scale factor $\frac{1}{T_0}$

This gives

$$\sum_{n=-\infty}^{\infty} e^{-j\omega n T_0} = \frac{2\pi}{T_0} \sum_{k=-\infty}^{\infty} \delta(\omega - k\omega_0), \qquad \omega_0 = \frac{2\pi}{T_0}$$

Hence, the Fourier transform of a **train of impulses spaced T_0 apart in time** is a **train of impulses spaced $\frac{2\pi}{T_0}$ apart in (angular) frequency**, scaled by $\frac{2\pi}{T_0}$

*If the derivation is done using f rather than ω, the result would be an impulse train spaced $\frac{1}{T_0}$ apart, scaled by $\frac{1}{T_0}$